

Software Requirements Specification (SRS)

1. Introduction

1.1 Purpose

This document defines the Software Requirements Specification (SRS) for a **Photo Frame Making Website**. The system will allow users to create a digital photo card by uploading an image, adding personal details (name and designation), choosing image and text placement, downloading the final image, and sharing it on social media platforms.

1.2 Scope

The application is a web-based platform designed for individual users and organizations to quickly generate branded or personalized photo frames/cards. The system will support image upload, customization, rendering, export, and social media sharing, with a focus on usability, performance, and responsiveness.

1.3 Definitions, Acronyms, and Abbreviations

- **SRS**: Software Requirements Specification
- **UI**: User Interface
- **UX**: User Experience
- **API**: Application Programming Interface
- **PNG/JPEG**: Image file formats
- **CDN**: Content Delivery Network

1.4 References

- IEEE 830 / IEEE 29148 SRS Standards
 - HTML5, CSS3, JavaScript Specifications
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2. Overall Description

2.1 Product Perspective

The Photo Frame Making Website is a standalone web application that runs on modern browsers. It does not require installation and is accessible via desktop and mobile devices.

2.2 Product Functions

- Upload user image
- Select predefined photo card/frame templates
- Add and edit name and designation text
- Adjust image placement (scale, position, alignment)

- Preview the final photo card
- Download the generated image
- Share the image on social media platforms

2.3 User Classes and Characteristics

- **General Users:** Non-technical users creating photo cards for personal or social use
- **Corporate Users:** Organizations creating branded frames for campaigns or events
- **Admin (Optional):** Manages templates and system configurations

2.4 Operating Environment

- Client: Modern web browsers (Chrome, Firefox, Safari, Edge)
- Server: Cloud-based hosting (AWS, GCP, or similar)
- Supported Devices: Desktop, tablet, mobile

2.5 Design and Implementation Constraints

- Browser-based image rendering limitations
- Image file size restrictions
- Cross-browser compatibility

2.6 Assumptions and Dependencies

- Users have stable internet connectivity
 - Social media APIs are accessible and functional
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3. Functional Requirements

3.1 User Interface Requirements

- Clean and intuitive UI
- Responsive layout for all screen sizes
- Real-time preview of changes

3.2 Image Upload Module

- Users can upload images in JPG, PNG formats
- Maximum image size: configurable (e.g., 5 MB)
- Client-side validation for file type and size

3.3 Photo Frame / Card Template Selection

- System shall provide predefined frame templates
- Templates may include placeholders for image, name, and designation

3.4 Text Input and Customization

- Users can input name and designation
- Editable font size, color, and alignment (basic level)
- Live preview of text changes

3.5 Image Placement and Adjustment

- Users can resize and reposition uploaded images
- Drag-and-drop functionality within defined boundaries
- Maintain aspect ratio option

3.6 Preview Functionality

- System shall display a real-time preview of the final photo card

3.7 Image Generation and Download

- System shall generate a high-quality image (PNG/JPEG)
- Users can download the generated image to their device

3.8 Social Media Sharing

- Share options for platforms such as Facebook, LinkedIn, X (Twitter), and WhatsApp
- Generate a shareable image or link
- Use platform-supported sharing APIs

3.9 Error Handling

- Display user-friendly error messages
- Handle invalid uploads and rendering failures gracefully

4. Non-Functional Requirements

4.1 Performance Requirements

- Image rendering time should be less than 3 seconds under normal load
- System should support concurrent users

4.2 Security Requirements

- No permanent storage of user images unless explicitly required
- Secure image upload handling
- Protection against malicious file uploads

4.3 Usability Requirements

- Minimal steps to complete photo card creation

- Clear call-to-action buttons

4.4 Reliability and Availability

- System uptime target: 99.5%
- Graceful degradation during high traffic

4.5 Scalability

- Architecture should support future addition of templates and features
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5. System Architecture (High-Level)

5.1 Frontend

- HTML5, CSS3, JavaScript (React/Vue optional)
- Canvas API or similar for image rendering

5.2 Backend

- Node.js / PHP / Python-based API
- Image processing service (server-side if required)

5.3 Storage

- Temporary storage for uploaded images
 - CDN for static assets
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6. Future Enhancements

- User accounts and saved designs
 - Advanced text styling and filters
 - AI-based auto-cropping and alignment
 - Admin dashboard for template management
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7. Approval and Sign-Off

This document serves as the baseline for system design, development, and testing of the Photo Frame Making Website. Any changes must follow a formal change management process.